

23-24-SIM-Midterm Template

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1 Class attendance Data

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Load `attendraw.RData` file in your current R or RStudio session. Selected numeric target is `stndfnl` (standardized final exam score) and let `freshman` status be the qualitative target. Prepare your dataset to represent factors in a suitable way.

```
## null device
##           1
```

All questions account for 1 point (you have to answer all of them)

1. Determine thresholds for mild and severe outliers for the target. Are there any outliers? Indicate observation id's and atypical values. *Do not take any action.*

```
Q3 <- quantile(df$stndfnl, 0.75)
Q1 <- quantile(df$stndfnl, 0.25)
IQR <- Q3 - Q1
mild_up <- Q3 + 1.5*IQR
mild_bot <- Q1 - 1.5*IQR
```

```
sev_up <- Q3 + 3*IQR
sev_bot <- Q1 - 3*IQR
```

```
mild_ids <- which(((df$stndfnl > mild_up) & (df$stndfnl < sev_up)) | ((df$stndfnl < mild_bot) & (df$stndfnl > sev_bot)))
sev_ids <- which((df$stndfnl > sev_up) | (df$stndfnl < sev_bot))
```

2. Use an imputation method to address missing data in the dataset. Validate imputation results.

The only variable containing null values is `hwrt` which contains 6. After performing an imputation with PCA we see that the quartile distribution and the mean remain almost intact, therefore we can safely apply this imputation.

```
summary(df)
```

	attend	termgpa	priGPA	ACT
## Min.	: 2.00	Min. :0.000	Min. :0.857	Min. :13.00
## 1st Qu.	:24.00	1st Qu.:2.138	1st Qu.:2.190	1st Qu.:20.00
## Median	:28.00	Median :2.670	Median :2.560	Median :22.00
## Mean	:26.15	Mean :2.601	Mean :2.587	Mean :22.51
## 3rd Qu.	:30.00	3rd Qu.:3.120	3rd Qu.:2.942	3rd Qu.:25.00

```
## Max. :32.00 Max. :4.000 Max. :3.930 Max. :32.00
##
## final atndrte hwrte frosh
## Min. :10.00 Min. : 6.25 Min. : 12.50 Freshman No :522
## 1st Qu.:22.00 1st Qu.: 75.00 1st Qu.: 87.50 Freshman Yes:158
## Median :26.00 Median : 87.50 Median :100.00
## Mean :25.89 Mean : 81.71 Mean : 87.91
## 3rd Qu.:29.00 3rd Qu.: 93.75 3rd Qu.:100.00
## Max. :39.00 Max. :100.00 Max. :100.00
## NA's :6
## soph skipped stndfnl
## Sophomore No :288 Min. : 0.000 Min. : -3.30882
## Sophomore Yes:392 1st Qu.: 2.000 1st Qu.: -0.78782
## Median : 4.000 Median : 0.05252
## Mean : 5.853 Mean : 0.02966
## 3rd Qu.: 8.000 3rd Qu.: 0.68277
## Max. :30.000 Max. : 2.78361
##
```

```
library(missMDA)
res_pca <- imputePCA(num_vars)

summary(df$hwrte)
```

```
## Min. 1st Qu. Median Mean 3rd Qu. Max. NA's
## 12.50 87.50 100.00 87.91 100.00 100.00 6
```

```
summary(res_pca$completeObs[, "hwrte"])
```

```
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 12.50 87.50 100.00 87.57 100.00 100.00
```

```
miss_ids <- which(is.na(df$hwrte))
df[miss_ids, "hwrte"] <- res_pca$completeObs[miss_ids, "hwrte"]
```

3. Are there any multivariate outliers in the dataset? Indicate how many and which at 99% confidence. Explain what they seem to share in common. Do not take any action.

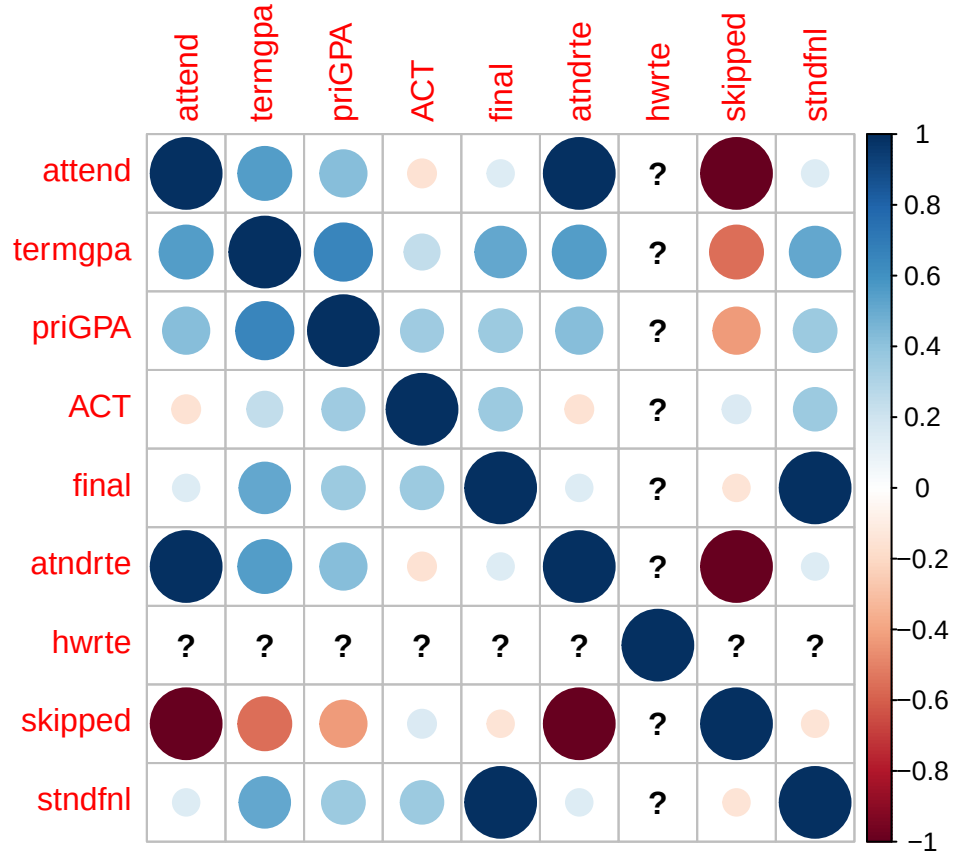
We see that attend, atndrte and skipped have perfect correlations (direct and inverse respectively), similarly with final and stndfnl. This is due to these variables being derived from the same data, therefore we will only keep stndfnl and attend to avoid the high correlations impeding us from computing multivariate outliers.

The most associated vars that define outliers are skipped, attend, atndrte, hwrte. One key characteristic of outliers is that they have a higher skipped mean, a significantly lower attend and atndrte and a dramatically lower hwrte (mean). We can also appreciate how their priGPA and termgpa are also lower.

```
library(corrplot)
```

```
## corrplot 0.95 loaded
```

```
corrplot(cor(num_vars))
```



```
num_vars <- df[sapply(df, is.numeric)]
num_vars[, c("attend", "skipped", "stndfni", "hwrte")] <- NULL
cat_vars <- df[!sapply(df, is.numeric)]
```

```
num_vars
```

```
##      termgpa priGPA ACT final atndrte
## 1      3.190  2.640  23    28  84.375
## 2      2.730  3.520  25    26  68.750
## 3      3.000  2.460  24    30  93.750
## 4      2.040  2.610  20    27  96.875
## 5      3.680  3.320  23    34 100.000
## 6      3.230  2.930  26    25  90.625
## 7      1.540  1.940  21    10  93.750
## 8      2.000  2.120  22    34  81.250
## 9      2.250  2.060  24    26  75.000
## 10     3.000  2.730  21    26  90.625
## 11     3.210  2.290  16    27  93.750
## 12     2.050  2.290  24    22  59.375
## 13     2.130  2.640  24    27  59.375
## 14     3.580  3.390  27    33  90.625
## 15     2.500  2.760  25    29  90.625
## 16     2.100  2.330  20    23  93.750
## 17     1.270  1.980  22    22  62.500
## 18     2.770  2.290  25    21 100.000
## 19     3.150  3.020  18    25  96.875
```

## 20	2.380	2.330	22	22	90.625
## 21	1.770	2.540	21	24	84.375
## 22	2.330	2.390	23	27	78.125
## 23	2.330	2.430	21	19	96.875
## 24	3.880	3.880	28	33	68.750
## 25	3.290	2.580	26	33	84.375
## 26	3.800	3.360	29	31	90.625
## 27	3.500	3.020	28	32	100.000
## 28	2.140	2.120	16	22	78.125
## 29	2.880	2.370	23	30	93.750
## 30	3.270	2.930	22	27	78.125
## 31	3.290	2.020	28	30	84.375
## 32	2.560	2.780	21	26	84.375
## 33	3.250	2.850	29	25	71.875
## 34	2.710	2.280	22	28	100.000
## 35	2.860	3.460	16	27	96.875
## 36	2.500	2.630	25	23	78.125
## 37	2.150	2.190	22	32	56.250
## 38	3.500	3.570	21	32	100.000
## 39	2.160	1.980	18	26	96.875
## 40	2.400	3.050	26	19	75.000
## 41	3.540	3.450	25	31	96.875
## 42	3.650	2.630	23	31	96.875
## 43	2.500	2.620	24	19	100.000
## 44	2.120	2.130	18	24	78.125
## 45	2.500	2.420	18	26	96.875
## 46	3.810	3.430	29	34	100.000
## 47	3.100	2.350	21	19	93.750
## 48	3.140	3.610	26	24	93.750
## 49	3.000	2.860	24	29	93.750
## 50	1.210	2.050	30	29	31.250
## 51	2.180	2.950	29	23	59.375
## 52	2.810	2.410	19	20	93.750
## 53	2.100	2.320	23	27	53.125
## 54	2.070	2.300	21	25	84.375
## 55	1.360	1.790	21	22	84.375
## 56	3.000	2.300	21	30	84.375
## 57	1.960	2.480	20	21	87.500
## 58	2.860	2.870	22	35	93.750
## 59	2.610	2.020	25	24	93.750
## 60	2.710	2.660	21	23	93.750
## 61	2.890	2.810	25	23	96.875
## 62	2.860	2.340	21	28	81.250
## 63	2.350	2.530	27	31	78.125
## 64	2.640	2.090	19	21	75.000
## 65	3.290	2.670	21	35	59.375
## 66	1.540	1.900	25	18	68.750
## 67	3.320	2.800	21	21	87.500
## 68	4.000	3.930	25	35	100.000
## 69	2.630	2.550	26	25	90.625
## 70	2.380	2.170	21	22	59.375
## 71	1.300	1.580	18	21	62.500
## 72	2.760	2.120	20	29	90.625
## 73	2.330	2.200	14	20	81.250

## 74	2.640	2.020	27	29	81.250
## 75	2.310	3.450	23	25	71.875
## 76	2.230	1.960	20	20	84.375
## 77	2.310	2.780	23	24	96.875
## 78	3.880	3.590	27	35	96.875
## 79	2.380	2.000	15	19	90.625
## 80	3.220	3.590	27	25	96.875
## 81	2.260	2.040	25	28	75.000
## 82	0.750	2.120	20	22	78.125
## 83	2.850	3.350	20	28	93.750
## 84	1.650	2.410	19	25	100.000
## 85	2.920	2.560	16	20	93.750
## 86	2.500	2.360	20	26	71.875
## 87	2.140	2.590	25	25	75.000
## 88	3.220	3.170	25	28	100.000
## 89	2.390	2.610	19	23	62.500
## 90	3.150	2.950	18	27	81.250
## 91	1.750	2.200	20	26	84.375
## 92	1.750	2.520	28	25	75.000
## 93	2.810	2.550	22	25	93.750
## 94	3.050	2.900	25	27	96.875
## 95	2.270	2.530	26	21	71.875
## 96	2.920	2.270	19	28	78.125
## 97	2.500	2.710	23	26	90.625
## 98	3.710	2.000	26	34	87.500
## 99	1.500	2.290	21	21	62.500
## 100	2.110	1.960	24	22	96.875
## 101	3.280	2.460	18	34	96.875
## 102	1.600	2.180	20	27	71.875
## 103	3.330	3.050	23	30	78.125
## 104	2.040	2.470	22	23	87.500
## 105	2.860	2.760	25	27	90.625
## 106	0.821	1.750	17	22	53.125
## 107	2.460	2.740	21	30	87.500
## 108	4.000	3.070	19	35	100.000
## 109	1.360	1.910	22	21	62.500
## 110	0.000	2.250	25	20	6.250
## 111	1.830	2.030	17	22	65.625
## 112	3.750	3.060	30	33	87.500
## 113	1.650	2.260	17	30	81.250
## 114	3.000	3.190	28	20	75.000
## 115	2.670	2.570	20	20	96.875
## 116	2.870	2.740	17	30	90.625
## 117	2.310	2.170	22	23	65.625
## 118	2.630	3.300	26	25	75.000
## 119	1.830	2.530	27	27	62.500
## 120	2.000	2.070	23	19	93.750
## 121	3.590	3.250	25	30	96.875
## 122	2.880	2.820	23	19	100.000
## 123	1.930	2.450	20	31	71.875
## 124	2.700	3.020	20	28	93.750
## 125	2.690	2.840	23	21	87.500
## 126	3.380	3.390	23	25	78.125
## 127	2.500	0.857	18	30	87.500

## 128	3.530	2.310	32	32	75.000
## 129	2.150	2.960	16	20	81.250
## 130	2.680	2.380	21	23	78.125
## 131	2.710	2.150	20	24	84.375
## 132	2.620	1.900	23	29	71.875
## 133	2.190	1.800	19	19	90.625
## 134	2.680	1.740	21	29	68.750
## 135	2.750	2.220	21	26	87.500
## 136	1.600	2.490	21	19	78.125
## 137	2.380	2.920	20	20	78.125
## 138	2.300	1.980	22	27	84.375
## 139	3.360	1.830	17	30	71.875
## 140	3.250	3.280	21	26	100.000
## 141	1.500	1.980	16	23	84.375
## 142	2.960	3.020	19	25	96.875
## 143	3.100	2.870	26	20	78.125
## 144	2.040	2.750	19	26	81.250
## 145	3.210	3.450	30	31	90.625
## 146	3.120	3.210	24	22	87.500
## 147	1.500	2.430	18	32	93.750
## 148	1.960	2.560	23	16	90.625
## 149	3.340	3.420	22	24	90.625
## 150	2.900	2.430	21	22	87.500
## 151	3.650	3.490	23	31	100.000
## 152	3.030	2.340	22	25	78.125
## 153	2.680	2.480	28	27	87.500
## 154	2.430	3.210	23	26	87.500
## 155	1.500	2.100	17	24	75.000
## 156	2.970	2.820	25	28	90.625
## 157	3.710	3.220	27	37	93.750
## 158	2.270	2.100	19	21	90.625
## 159	2.680	1.670	22	31	87.500
## 160	3.630	3.640	26	32	78.125
## 161	2.590	2.450	25	29	93.750
## 162	2.770	2.530	20	29	100.000
## 163	3.300	2.390	16	27	90.625
## 164	2.130	2.630	20	23	87.500
## 165	2.350	2.540	24	30	75.000
## 166	3.460	3.540	27	29	81.250
## 167	2.080	1.850	22	23	9.375
## 168	3.050	2.440	24	32	46.875
## 169	3.550	3.310	22	28	96.875
## 170	2.130	2.350	20	30	78.125
## 171	2.380	2.220	20	22	84.375
## 172	3.730	3.200	25	37	84.375
## 173	2.130	2.580	21	17	84.375
## 174	3.710	3.610	26	31	90.625
## 175	2.550	2.460	18	24	100.000
## 176	3.130	3.340	28	26	71.875
## 177	3.190	3.130	17	23	100.000
## 178	2.590	2.790	26	28	87.500
## 179	2.790	2.360	23	27	96.875
## 180	3.000	3.290	28	30	78.125
## 181	2.750	2.370	20	24	93.750

## 182	3.250	3.570	29	30	75.000
## 183	0.700	2.070	28	28	68.750
## 184	3.210	3.200	27	30	96.875
## 185	1.810	1.980	23	25	50.000
## 186	3.210	2.800	22	22	87.500
## 187	3.000	2.980	18	22	96.875
## 188	1.850	2.850	22	21	65.625
## 189	2.000	1.950	20	26	28.125
## 190	1.350	1.890	21	22	75.000
## 191	3.040	3.290	19	32	96.875
## 192	3.180	2.760	25	26	87.500
## 193	3.360	1.780	23	34	81.250
## 194	3.000	2.890	20	26	96.875
## 195	2.880	1.950	25	26	62.500
## 196	3.040	2.480	27	30	71.875
## 197	2.730	2.300	16	24	90.625
## 198	2.780	1.860	21	34	100.000
## 199	2.960	2.700	24	30	100.000
## 200	2.250	2.280	15	26	90.625
## 201	2.110	2.690	19	19	87.500
## 202	3.270	2.470	25	34	96.875
## 203	2.230	2.670	19	22	87.500
## 204	2.940	2.400	22	22	96.875
## 205	4.000	3.450	30	36	93.750
## 206	2.320	2.350	21	27	84.375
## 207	3.410	2.920	23	24	78.125
## 208	2.290	2.730	18	26	84.375
## 209	1.540	2.680	19	22	78.125
## 210	3.560	3.520	23	31	100.000
## 211	1.000	1.410	23	20	31.250
## 212	2.940	2.940	20	30	78.125
## 213	3.140	3.320	23	22	96.875
## 214	2.750	2.910	19	22	90.625
## 215	2.230	2.130	24	26	84.375
## 216	3.800	3.110	24	29	93.750
## 217	3.540	2.940	16	22	100.000
## 218	3.750	3.530	26	31	96.875
## 219	3.500	2.830	22	27	100.000
## 220	2.620	2.660	26	23	90.625
## 221	1.760	2.120	23	26	75.000
## 222	2.360	2.200	19	22	65.625
## 223	2.460	2.250	26	24	65.625
## 224	3.730	3.220	28	32	100.000
## 225	3.230	2.950	21	26	87.500
## 226	2.610	2.660	17	31	100.000
## 227	2.630	2.410	20	25	87.500
## 228	3.080	2.590	19	27	87.500
## 229	1.380	2.550	19	17	93.750
## 230	3.540	3.480	28	29	93.750
## 231	3.150	2.910	23	26	87.500
## 232	2.210	2.280	18	26	90.625
## 233	2.180	2.490	30	24	53.125
## 234	3.100	2.940	23	25	100.000
## 235	2.620	3.110	27	24	81.250

## 236	2.400	2.690	23	28	96.875
## 237	3.600	3.360	28	21	93.750
## 238	2.040	2.120	22	32	81.250
## 239	3.120	2.410	23	28	90.625
## 240	3.190	2.300	19	22	87.500
## 241	1.970	2.070	24	24	78.125
## 242	3.000	3.180	25	33	100.000
## 243	3.360	2.620	27	28	96.875
## 244	1.790	1.740	24	25	75.000
## 245	2.970	3.150	28	27	84.375
## 246	2.670	2.310	22	27	71.875
## 247	3.160	2.840	20	26	90.625
## 248	2.500	2.070	23	22	81.250
## 249	1.710	2.130	20	21	68.750
## 250	2.730	2.550	24	27	90.625
## 251	3.580	2.790	28	30	90.625
## 252	2.210	2.390	20	26	84.375
## 253	1.750	2.190	22	21	56.250
## 254	2.110	2.140	20	23	84.375
## 255	4.000	3.840	28	37	90.625
## 256	3.140	3.210	22	24	71.875
## 257	2.920	2.680	20	27	78.125
## 258	2.290	2.460	19	19	87.500
## 259	2.960	2.340	24	25	93.750
## 260	2.070	2.620	25	21	75.000
## 261	1.800	2.610	20	30	93.750
## 262	3.730	3.810	32	38	100.000
## 263	2.310	2.420	18	19	87.500
## 264	2.390	2.450	20	25	90.625
## 265	2.960	2.930	28	35	78.125
## 266	3.620	3.480	24	32	96.875
## 267	3.120	2.960	22	29	96.875
## 268	2.310	2.360	19	30	93.750
## 269	1.710	2.890	24	22	71.875
## 270	2.690	3.020	26	29	96.875
## 271	2.430	2.590	16	21	100.000
## 272	2.550	2.570	15	20	100.000
## 273	2.770	2.530	24	29	93.750
## 274	3.400	3.660	29	26	87.500
## 275	2.500	2.100	17	21	81.250
## 276	3.000	1.810	22	30	75.000
## 277	2.250	2.390	16	19	87.500
## 278	2.500	2.030	26	24	46.875
## 279	2.050	2.590	22	24	87.500
## 280	1.380	1.920	16	16	90.625
## 281	2.730	1.850	23	25	84.375
## 282	3.300	3.540	26	33	87.500
## 283	2.790	2.570	19	31	96.875
## 284	3.310	3.270	26	27	90.625
## 285	1.790	2.440	20	14	93.750
## 286	2.970	2.370	22	31	75.000
## 287	2.130	2.630	19	14	96.875
## 288	3.040	2.680	25	24	87.500
## 289	2.040	2.500	18	23	65.625

## 290	2.750	2.950	21	25	81.250
## 291	3.400	3.160	25	29	100.000
## 292	2.820	3.130	22	19	100.000
## 293	0.545	1.980	18	23	65.625
## 294	1.430	2.030	28	25	28.125
## 295	2.150	1.930	16	27	81.250
## 296	2.540	1.880	23	24	75.000
## 297	2.500	2.860	23	25	81.250
## 298	2.380	2.300	17	25	96.875
## 299	1.670	2.190	13	15	84.375
## 300	3.210	2.540	23	25	87.500
## 301	2.750	2.940	22	24	100.000
## 302	2.500	2.930	25	29	78.125
## 303	2.900	2.600	19	24	81.250
## 304	2.650	3.260	20	29	84.375
## 305	3.430	3.290	20	22	100.000
## 306	2.850	2.190	24	31	65.625
## 307	3.210	3.020	27	18	100.000
## 308	2.250	2.090	20	25	78.125
## 309	2.440	2.110	24	22	87.500
## 310	2.770	2.830	24	32	75.000
## 311	3.400	2.350	21	27	71.875
## 312	3.460	2.380	23	30	96.875
## 313	3.340	3.140	25	27	100.000
## 314	2.430	2.480	22	22	90.625
## 315	2.710	2.520	21	22	90.625
## 316	2.110	2.760	24	26	96.875
## 317	4.000	3.700	24	39	90.625
## 318	2.100	2.080	22	30	75.000
## 319	2.380	2.390	22	24	53.125
## 320	2.120	2.460	21	18	90.625
## 321	4.000	3.700	25	35	96.875
## 322	3.040	1.990	25	26	84.375
## 323	1.110	2.580	25	28	62.500
## 324	3.130	2.670	21	30	96.875
## 325	3.000	1.910	23	29	84.375
## 326	1.800	1.960	24	25	53.125
## 327	1.680	2.150	20	24	53.125
## 328	3.000	2.940	22	24	90.625
## 329	2.250	2.110	15	25	84.375
## 330	3.250	3.500	25	30	96.875
## 331	1.140	1.840	22	17	37.500
## 332	2.970	3.100	24	25	84.375
## 333	1.890	2.800	23	25	78.125
## 334	2.670	1.760	23	26	84.375
## 335	3.200	2.970	22	29	81.250
## 336	2.500	2.370	23	23	84.375
## 337	1.790	1.680	21	19	71.875
## 338	2.000	2.270	17	22	78.125
## 339	3.620	3.100	28	32	84.375
## 340	2.500	2.380	16	28	87.500
## 341	3.050	2.840	15	24	100.000
## 342	3.640	3.690	26	31	87.500
## 343	3.540	3.370	25	31	81.250

## 344	2.860	2.170	23	28	71.875
## 345	3.680	3.600	23	31	100.000
## 346	3.230	3.280	26	29	87.500
## 347	3.150	3.490	25	18	93.750
## 348	3.110	2.840	21	31	84.375
## 349	2.590	2.920	24	28	78.125
## 350	2.790	2.750	24	25	100.000
## 351	3.040	3.170	24	26	78.125
## 352	2.180	2.440	22	22	84.375
## 353	2.810	2.350	22	22	87.500
## 354	2.900	2.190	18	24	100.000
## 355	3.500	3.280	28	33	93.750
## 356	2.170	2.250	25	36	56.250
## 357	1.350	1.590	31	21	46.875
## 358	0.792	1.650	23	23	65.625
## 359	3.190	2.620	19	34	75.000
## 360	3.380	3.290	23	29	100.000
## 361	1.850	1.430	25	27	56.250
## 362	2.630	3.390	24	27	75.000
## 363	2.040	2.390	21	22	78.125
## 364	4.000	2.240	25	35	93.750
## 365	3.860	3.870	31	34	100.000
## 366	3.070	2.900	18	23	100.000
## 367	2.650	2.640	23	19	90.625
## 368	2.500	2.250	31	35	31.250
## 369	3.630	2.180	23	29	84.375
## 370	1.630	2.460	23	24	59.375
## 371	2.650	2.670	28	26	53.125
## 372	2.810	3.250	22	30	90.625
## 373	2.380	2.430	23	26	81.250
## 374	1.650	2.230	20	22	50.000
## 375	2.670	2.630	24	24	56.250
## 376	1.680	1.980	22	13	53.125
## 377	2.310	1.800	27	28	18.750
## 378	3.000	2.810	21	30	84.375
## 379	2.640	2.270	21	22	75.000
## 380	2.000	2.770	22	19	81.250
## 381	1.620	2.080	26	19	46.875
## 382	2.690	2.500	25	28	87.500
## 383	3.950	3.810	31	36	93.750
## 384	1.620	2.570	21	25	90.625
## 385	2.430	1.930	20	17	93.750
## 386	2.500	2.510	24	28	87.500
## 387	3.460	2.000	20	35	87.500
## 388	3.090	2.590	23	33	100.000
## 389	2.330	2.610	24	19	71.875
## 390	1.230	2.510	25	21	37.500
## 391	3.250	2.150	27	32	81.250
## 392	1.770	1.910	27	25	15.625
## 393	4.000	3.920	29	37	93.750
## 394	2.630	2.840	25	22	78.125
## 395	3.700	3.310	24	31	84.375
## 396	1.960	2.040	21	17	87.500
## 397	1.170	1.840	19	28	43.750

## 398	2.880	2.970	24	27	87.500
## 399	2.330	2.360	18	22	87.500
## 400	2.000	2.150	19	17	71.875
## 401	2.150	2.000	20	23	87.500
## 402	1.880	1.740	15	27	84.375
## 403	2.900	2.700	21	23	100.000
## 404	3.500	3.590	28	35	90.625
## 405	2.410	2.750	21	23	78.125
## 406	2.380	2.740	19	25	90.625
## 407	3.040	2.480	16	24	68.750
## 408	2.410	2.480	27	25	96.875
## 409	2.450	2.440	18	23	90.625
## 410	3.660	2.860	27	30	81.250
## 411	1.530	2.020	28	32	31.250
## 412	1.890	1.970	21	21	84.375
## 413	3.700	2.820	24	34	71.875
## 414	4.000	3.900	28	38	100.000
## 415	2.330	2.420	19	17	87.500
## 416	2.850	2.500	25	31	75.000
## 417	2.000	2.130	15	26	75.000
## 418	3.620	3.800	27	31	93.750
## 419	2.400	3.020	23	27	96.875
## 420	3.710	3.500	28	35	93.750
## 421	3.540	3.320	23	32	100.000
## 422	3.570	2.840	19	26	100.000
## 423	3.110	2.740	19	24	96.875
## 424	2.130	2.030	22	19	87.500
## 425	3.500	3.230	24	28	96.875
## 426	3.310	2.190	27	26	84.375
## 427	2.250	2.240	21	24	71.875
## 428	2.380	2.150	23	28	78.125
## 429	1.460	2.320	25	32	87.500
## 430	3.040	2.940	19	27	87.500
## 431	2.680	2.700	23	21	71.875
## 432	1.110	1.200	25	18	43.750
## 433	0.692	1.740	21	19	84.375
## 434	2.960	2.800	16	23	90.625
## 435	2.370	2.460	23	32	78.125
## 436	2.250	2.200	19	24	78.125
## 437	2.250	2.600	26	31	84.375
## 438	2.500	2.210	16	29	93.750
## 439	2.000	2.170	18	21	87.500
## 440	3.000	2.600	23	23	81.250
## 441	2.130	2.940	28	29	18.750
## 442	1.230	2.710	25	26	84.375
## 443	2.000	1.800	16	26	81.250
## 444	3.500	2.590	19	29	87.500
## 445	2.000	2.640	24	24	53.125
## 446	2.320	2.100	25	30	75.000
## 447	1.900	2.490	24	20	81.250
## 448	2.500	2.430	22	19	93.750
## 449	3.290	2.900	25	35	78.125
## 450	2.000	2.910	23	20	90.625
## 451	3.410	2.980	27	35	90.625

## 452	1.500	2.390	19	24	65.625
## 453	3.500	3.040	27	28	90.625
## 454	0.567	1.740	21	23	59.375
## 455	3.190	3.500	25	32	90.625
## 456	1.880	2.540	17	20	87.500
## 457	2.000	2.450	17	29	100.000
## 458	2.620	2.980	31	22	81.250
## 459	2.820	1.960	22	26	90.625
## 460	3.140	2.020	20	20	87.500
## 461	2.850	1.910	20	27	100.000
## 462	2.130	2.300	17	20	96.875
## 463	2.640	2.560	19	19	84.375
## 464	2.120	2.210	20	19	84.375
## 465	3.350	2.120	17	29	75.000
## 466	2.730	3.090	25	22	81.250
## 467	3.000	1.500	20	22	96.875
## 468	1.750	1.980	17	22	62.500
## 469	2.880	2.820	25	21	87.500
## 470	3.230	3.120	28	30	71.875
## 471	2.960	2.680	21	24	96.875
## 472	2.320	2.580	23	20	37.500
## 473	2.540	2.260	21	24	90.625
## 474	0.333	2.470	26	23	37.500
## 475	1.820	2.260	18	17	84.375
## 476	3.340	3.210	21	28	96.875
## 477	2.900	2.660	22	29	90.625
## 478	2.920	3.000	23	25	81.250
## 479	3.160	3.250	23	26	90.625
## 480	3.050	2.830	21	19	93.750
## 481	2.540	2.840	22	33	68.750
## 482	2.100	1.950	22	30	90.625
## 483	3.770	3.530	25	27	93.750
## 484	1.250	1.750	27	28	46.875
## 485	2.650	2.270	25	17	90.625
## 486	2.300	2.040	19	21	78.125
## 487	1.070	1.320	21	16	90.625
## 488	2.560	2.710	24	24	78.125
## 489	2.540	1.880	22	30	65.625
## 490	2.910	2.670	23	28	96.875
## 491	3.000	2.920	19	24	81.250
## 492	2.430	1.340	18	21	90.625
## 493	3.210	2.680	26	32	96.875
## 494	1.600	2.290	24	17	31.250
## 495	2.850	2.390	24	29	90.625
## 496	3.030	3.600	29	31	71.875
## 497	3.040	2.560	27	30	81.250
## 498	1.690	2.610	23	20	84.375
## 499	1.950	2.750	24	30	84.375
## 500	1.930	2.380	20	28	65.625
## 501	1.810	2.750	27	27	56.250
## 502	2.860	3.130	21	11	84.375
## 503	3.250	2.630	26	30	84.375
## 504	2.500	2.620	20	23	87.500
## 505	2.390	2.290	17	30	84.375

## 506	2.530	2.890	32	29	50.000
## 507	2.650	1.250	25	29	71.875
## 508	3.420	2.510	23	26	68.750
## 509	2.680	2.460	22	24	81.250
## 510	1.790	1.820	20	25	59.375
## 511	1.350	1.520	27	30	28.125
## 512	3.380	3.000	21	16	93.750
## 513	0.450	2.140	27	25	12.500
## 514	3.880	3.230	22	30	100.000
## 515	2.850	2.380	20	29	96.875
## 516	2.310	1.880	20	24	87.500
## 517	2.650	2.770	23	28	87.500
## 518	2.860	2.610	25	25	78.125
## 519	1.630	2.010	20	27	78.125
## 520	3.070	2.890	26	31	100.000
## 521	2.120	2.160	22	31	59.375
## 522	4.000	3.810	25	36	100.000
## 523	1.960	2.000	22	28	78.125
## 524	1.770	1.890	19	17	87.500
## 525	2.690	2.120	23	28	75.000
## 526	2.970	2.880	25	30	100.000
## 527	3.270	3.050	22	32	96.875
## 528	2.310	2.230	22	22	84.375
## 529	1.880	2.110	22	24	93.750
## 530	2.250	2.610	18	24	84.375
## 531	2.750	2.980	31	35	90.625
## 532	1.790	2.730	27	24	81.250
## 533	3.860	3.930	24	34	100.000
## 534	2.080	1.990	24	23	65.625
## 535	2.960	2.760	23	31	87.500
## 536	2.230	1.830	18	21	81.250
## 537	2.460	2.580	20	21	78.125
## 538	1.700	2.120	28	26	81.250
## 539	3.130	2.680	23	29	93.750
## 540	2.380	2.020	16	21	84.375
## 541	2.750	2.060	20	22	50.000
## 542	2.640	2.270	22	19	84.375
## 543	2.460	3.040	24	24	40.625
## 544	0.667	1.600	26	23	15.625
## 545	2.820	2.710	18	29	100.000
## 546	0.000	1.740	24	27	34.375
## 547	2.680	1.660	19	23	100.000
## 548	0.625	2.220	23	24	78.125
## 549	2.890	3.050	23	25	90.625
## 550	1.960	2.060	22	25	40.625
## 551	2.470	2.900	18	27	84.375
## 552	3.100	2.630	22	25	93.750
## 553	2.150	2.280	23	27	87.500
## 554	0.321	1.790	21	26	28.125
## 555	2.700	2.800	26	22	65.625
## 556	3.130	2.910	26	29	93.750
## 557	3.120	1.600	20	33	59.375
## 558	2.630	1.350	26	23	53.125
## 559	2.960	3.180	25	28	93.750

## 560	2.800	2.920	20	22	96.875
## 561	3.860	3.050	27	34	96.875
## 562	2.900	3.090	24	22	90.625
## 563	2.430	2.600	16	29	56.250
## 564	2.890	2.430	17	28	93.750
## 565	2.380	2.670	25	21	75.000
## 566	0.867	1.630	25	23	43.750
## 567	2.500	2.650	20	24	93.750
## 568	3.550	3.250	23	27	96.875
## 569	3.500	3.650	20	29	87.500
## 570	1.610	2.050	23	18	93.750
## 571	2.690	2.160	23	25	84.375
## 572	2.650	1.550	22	34	59.375
## 573	2.850	2.750	20	28	90.625
## 574	2.250	2.660	18	25	96.875
## 575	2.960	2.800	23	32	87.500
## 576	3.640	3.080	17	31	90.625
## 577	2.630	2.140	20	21	84.375
## 578	3.460	3.490	27	29	90.625
## 579	2.000	1.430	25	24	37.500
## 580	2.500	2.430	19	23	87.500
## 581	2.900	2.300	21	16	96.875
## 582	2.770	2.780	18	27	93.750
## 583	2.290	2.360	18	23	78.125
## 584	2.500	2.860	22	23	81.250
## 585	2.830	2.800	24	25	93.750
## 586	3.070	2.420	25	26	100.000
## 587	3.250	3.880	26	32	93.750
## 588	3.620	3.280	29	36	81.250
## 589	2.810	1.980	15	28	93.750
## 590	3.000	2.960	19	26	100.000
## 591	2.690	2.860	27	25	90.625
## 592	1.180	2.160	22	29	87.500
## 593	1.930	2.300	19	21	96.875
## 594	0.923	1.480	22	19	46.875
## 595	3.500	2.980	26	27	93.750
## 596	2.790	3.480	26	26	78.125
## 597	0.600	1.670	24	22	46.875
## 598	2.800	3.910	26	29	87.500
## 599	2.900	3.250	20	25	100.000
## 600	2.500	2.250	19	22	87.500
## 601	2.600	2.660	21	25	96.875
## 602	2.630	2.480	27	26	46.875
## 603	2.330	3.070	28	28	84.375
## 604	3.640	2.500	20	32	93.750
## 605	2.790	2.670	17	21	96.875
## 606	1.230	2.950	27	24	50.000
## 607	2.650	2.570	18	25	93.750
## 608	2.390	2.500	20	23	93.750
## 609	2.630	2.540	25	26	93.750
## 610	3.080	2.170	21	26	87.500
## 611	1.940	2.660	23	20	93.750
## 612	2.630	2.200	18	25	78.125
## 613	3.040	3.360	20	23	93.750

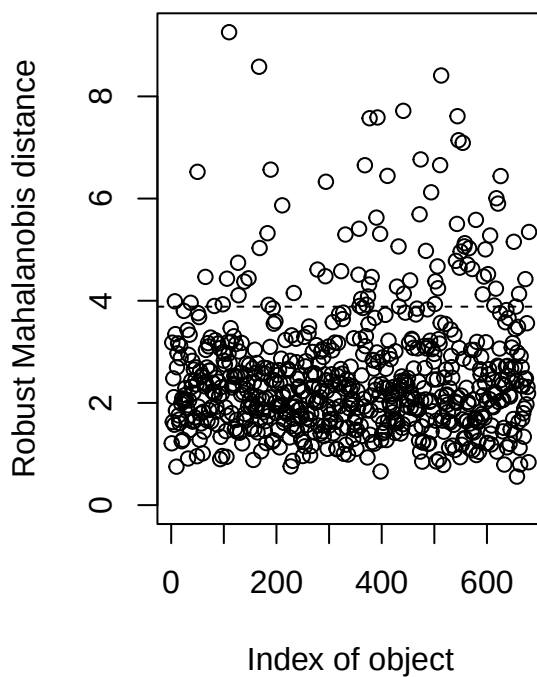
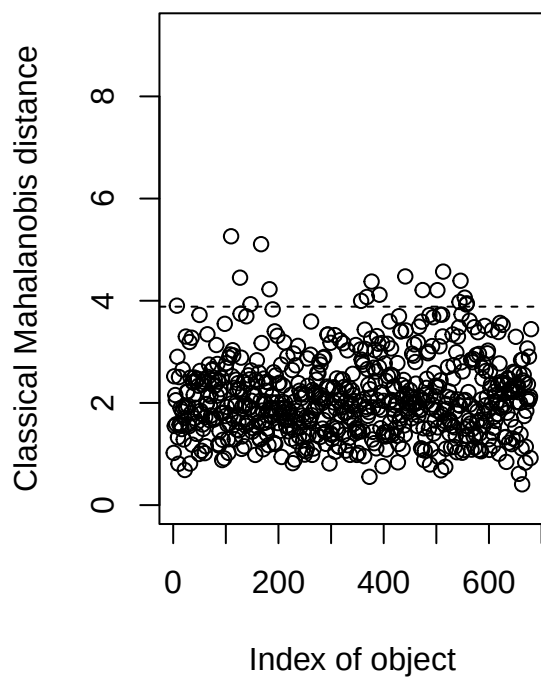
## 614	1.850	2.240	23	22	50.000
## 615	2.930	2.030	24	28	84.375
## 616	2.920	3.100	22	24	75.000
## 617	2.200	2.000	22	29	50.000
## 618	0.800	1.800	21	17	31.250
## 619	2.250	2.520	20	24	87.500
## 620	3.470	2.970	30	35	93.750
## 621	2.050	2.270	20	22	34.375
## 622	3.420	3.130	25	24	84.375
## 623	2.110	2.160	25	31	84.375
## 624	2.470	2.060	20	27	62.500
## 625	1.140	2.110	20	18	56.250
## 626	0.625	2.010	22	24	31.250
## 627	3.210	3.240	27	22	84.375
## 628	3.770	3.530	23	28	96.875
## 629	2.860	2.410	17	29	100.000
## 630	3.270	2.620	24	28	93.750
## 631	2.790	3.130	25	31	96.875
## 632	3.270	2.950	22	30	100.000
## 633	2.740	2.950	22	26	87.500
## 634	3.890	3.660	27	28	90.625
## 635	2.070	2.050	23	24	53.125
## 636	3.880	3.080	27	36	81.250
## 637	2.810	2.280	22	24	56.250
## 638	2.540	2.390	19	22	87.500
## 639	3.750	2.560	20	29	100.000
## 640	1.750	2.210	20	23	96.875
## 641	2.800	2.860	26	25	56.250
## 642	3.370	2.950	26	28	75.000
## 643	3.250	3.000	21	27	93.750
## 644	3.700	3.410	28	34	96.875
## 645	3.040	3.000	21	28	100.000
## 646	1.770	1.380	21	21	81.250
## 647	1.880	2.270	27	20	65.625
## 648	2.390	2.420	28	23	78.125
## 649	2.970	2.620	20	34	71.875
## 650	2.420	3.010	24	26	90.625
## 651	0.462	1.950	19	22	50.000
## 652	3.110	2.830	24	25	96.875
## 653	2.880	3.050	20	31	75.000
## 654	2.100	2.510	22	21	56.250
## 655	1.320	1.480	23	22	53.125
## 656	1.770	1.940	20	26	90.625
## 657	2.640	2.460	21	26	87.500
## 658	3.350	3.500	28	26	71.875
## 659	1.250	2.180	23	22	59.375
## 660	3.180	2.480	21	27	96.875
## 661	3.110	3.330	27	32	56.250
## 662	3.000	2.830	26	27	84.375
## 663	2.650	2.650	22	27	81.250
## 664	2.050	2.500	15	16	96.875
## 665	3.770	3.780	25	34	96.875
## 666	3.570	3.540	19	26	100.000
## 667	2.970	3.070	27	25	90.625

```
## 668 3.270 3.720 28 30 93.750
## 669 2.460 2.480 24 28 71.875
## 670 2.750 3.100 24 26 81.250
## 671 3.430 3.500 23 26 96.875
## 672 2.730 1.960 22 22 75.000
## 673 0.850 1.840 18 26 59.375
## 674 2.610 2.730 27 23 96.875
## 675 3.410 3.480 27 26 90.625
## 676 1.000 2.310 18 22 78.125
## 677 2.690 2.380 19 19 100.000
## 678 3.660 3.610 22 30 100.000
## 679 3.170 3.020 24 27 87.500
## 680 1.800 1.190 24 29 43.750
```

```
library(chemometrics)
```

```
## Cargando paquete requerido: rpart
```

```
res_mout <- Moutlier(num_vars, quantile=0.99)
```



```
mout_ids <- which((res_mout$md > res_mout$cutoff) & (res_mout$rd > res_mout$cutoff) )

plot(res_mout$md, res_mout$rd, main="Mahalanobis")
abline(h=res_mout$cutoff)
abline(v=res_mout$cutoff)

df$mout <- 0
df$mout[mout_ids] <- 1
```



```
df$mout <- factor(df$mout, labels=c("Mout NO", "Mout YES"))
```

```
library(FactoMineR)
```

```
res_cat <- catdes(df, num.var=12, proba=0.99)
```

```
res_cat$test.chi2
```

```
##          p.value df
```

```
## soph  0.3577411  1
```

```
## frosh 0.8192674  1
```

```
res_cat$quanti
```

```
## $`Mout NO`
```

	v.test	Mean	in category	Overall mean	sd	in category	Overall sd
atndrte	10.0580130	82.82431921	81.70955882	15.1381414	17.0344515		
attend	10.0580130	26.50378215	26.14705882	4.8442053	5.4510245		
hwrt	8.2675207	88.61863783	87.56792376	18.2328701	19.5329606		
termgpa	6.5259198	2.63225265	2.60100001	0.7026740	0.7360442		
priGPA	4.8876468	2.60408472	2.58677500	0.5351007	0.5443134		
final	0.9847587	25.92133132	25.89117647	4.6489322	4.7063704		
stndfnl	0.9847587	0.03599397	0.02965892	0.9766664	0.9887333		
ACT	-2.6199756	22.45083207	22.51029412	3.4587808	3.4882003		
skipped	-10.0580130	5.49621785	5.85294118	4.8442053	5.4510245		

```
##          p.value
```

```
## atndrte 8.469038e-24
```

```
## attend  8.469038e-24
```

```
## hwrt    1.367735e-16
```

```
## termgpa 6.758547e-11
```

```
## priGPA  1.020484e-06
```

```
## final   3.247426e-01
```

```
## stndfnl 3.247426e-01
```

```
## ACT     8.793606e-03
```

```
## skipped 8.469038e-24
```

```
##
```

```
## $`Mout YES`
```

	v.test	Mean	in category	Overall mean	sd	in category	Overall sd
skipped	10.0580130	18.2631579	5.85294118	9.4135519	5.4510245		
ACT	2.6199756	24.5789474	22.51029412	3.8568570	3.4882003		
stndfnl	-0.9847587	-0.1907342	0.02965892	1.3238947	0.9887333		
final	-0.9847587	24.8421053	25.89117647	6.3017388	4.7063704		
priGPA	-4.8876468	1.9845790	2.58677500	0.5188160	0.5443134		
termgpa	-6.5259198	1.5137368	2.60100001	0.9979461	0.7360442		
hwrt	-8.2675207	51.0141344	87.56792376	26.7409363	19.5329606		
atndrte	-10.0580130	42.9276316	81.70955882	29.4173495	17.0344515		
attend	-10.0580130	13.7368421	26.14705882	9.4135519	5.4510245		

```
##          p.value
```

```
## skipped 8.469038e-24
```

```
## ACT     8.793606e-03
```

```
## stndfnl 3.247426e-01
```

```
## final   3.247426e-01
```

```
## priGPA  1.020484e-06
```

```
## termgpa 6.758547e-11
```

```
## hwrt    1.367735e-16
```

```
## atndrte 8.469038e-24
```

```
## attend 8.469038e-24
```

```
res_cat$quanti.var
```

```
##           Eta2      P-value
## skipped 0.148989138 1.393199e-25
## attend  0.148989138 1.393199e-25
## atndrte 0.148989138 1.393199e-25
## hwrtte  0.100665535 2.282937e-17
## termgpa 0.062721104 3.482764e-11
## priGPA  0.035182756 8.391618e-07
## ACT     0.010109385 8.697220e-03
## final   0.001428203 3.251047e-01
## stndfnl 0.001428203 3.251047e-01
```

```
res_cat$category
```

```
## $`Mout NO`
```

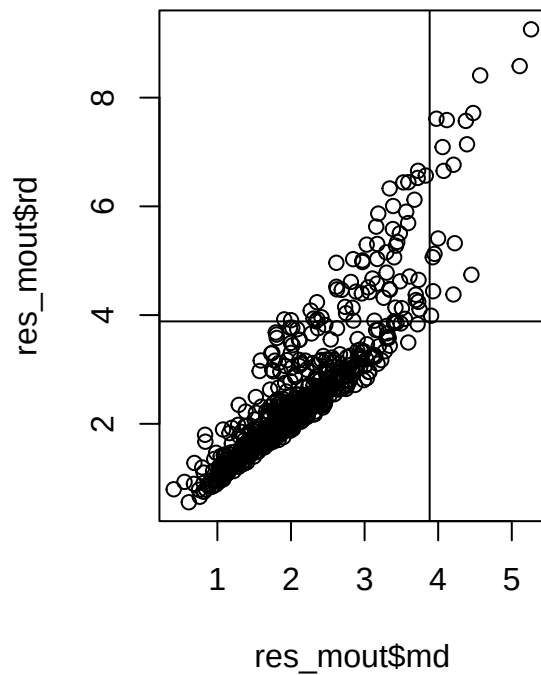
##		Cla/Mod	Mod/Cla	Global	p.value	v.test
##	soph=Sophomore Yes	97.70408	57.94251	57.64706	0.3699389	0.8965877
##	frosh=Freshman Yes	97.46835	23.29803	23.23529	0.8585764	0.1781866
##	frosh=Freshman No	97.12644	76.70197	76.76471	0.8585764	-0.1781866
##	soph=Sophomore No	96.52778	42.05749	42.35294	0.3699389	-0.8965877

```
##
```

```
## $`Mout YES`
```

##		Cla/Mod	Mod/Cla	Global	p.value	v.test
##	soph=Sophomore No	3.472222	52.63158	42.35294	0.3699389	0.8965877
##	frosh=Freshman No	2.873563	78.94737	76.76471	0.8585764	0.1781866
##	frosh=Freshman Yes	2.531646	21.05263	23.23529	0.8585764	-0.1781866
##	soph=Sophomore Yes	2.295918	47.36842	57.64706	0.3699389	-0.8965877

Mahalanobis



4. Analyze the profile of the numeric target (stndfnl) using a suitable profiling method. A detailed explanation of the procedure outcome is requested.

```
names(df)
```

```
## [1] "attend" "termgpa" "priGPA" "ACT" "final" "atndrte" "hwrt"
## [8] "frosh" "soph" "skipped" "stndfnl" "mout"
```

```
res_cond <- condres(df, num.var=11)
```

```
res_cond$quanti
```

```
## correlation p.value
## final 1.0000000 0.000000e+00
## termgpa 0.5106093 2.005168e-46
## priGPA 0.3659273 5.673134e-23
## ACT 0.3612486 2.170214e-22
## hwrt 0.1418869 2.056716e-04
## attend 0.1400327 2.493270e-04
## atndrte 0.1400327 2.493270e-04
## skipped -0.1400327 2.493270e-04
```

```
res_cond$quali
```

```
## R2 p.value
## frosh 0.007856995 0.02079203
```

```
res_cond$category
```

```
##               Estimate    p.value
## frosh=Freshman No   0.1037582 0.02079203
## frosh=Freshman Yes -0.1037582 0.02079203
```

5. Analyze the profile of the binary target (frosh) using a suitable method. A detailed explanation of the procedure outcome is requested.
6. Is there variance homogeneity in the stndfnl target groups defined by frosh and soph classes, one by one and when applied simultaneously? Hint: You have to define a new factor.
7. Mean stndfnl target can be considered to be the equal across groups defined by frosh target? Use a two.sided test at 1% significance level and indicate the confidence interval for freshman target population mean. Indicate whether equal variances and normal distribution of stndfnl hypothesis hold in the population.
8. State and test one.sided hypothesis to assess whether stndfnl is less for freshman than the rest at 1% significance level. Indicate and justify a 95% confidence interval for freshman target population mean.
9. The standard deviation of stndfnl in the freshman group should not exceed 1. For the sample in the freshman group in your dataset, calculate the deviation of stndfnl assuming that normal assumption holds. State and include any assumptions needed to test at the 1% significance level whether population standard deviation is larger than 1 in the freshman group. Figure out the 99% upper threshold for stndfnl in the freshman population standard deviation.
10. Determine a 99% confidence interval for the population proportion of neither freshman, nor sophomore. Test the null hypothesis that selecting a unit neither freshman, nor sophomore has a population probability greater than 0.25.
11. Test the null hypothesis that the proportion of freshman group and the population proportion of neither freshman, nor sophomore is the same at 1% significance level.

Do not forget to knit your .Rmd file to .pdf (or to word and afterwards to pdf) before posting it on the ATENEA platform task (only for pdf). Markdown should be also posted in the corresponding task.